

INDIAN EXPRESS UPSC IAS EDITION HD 22~06~2026

-:FOR UPSC IAS ASPIRANTS:-

"AVOID POLITICAL & IRRELEVANT ARTICLES"

Please Try To Read This Completely in 40 Minutes If You

Can't So Then You Have To Increase Your Efforts

All the topics of this UPSC IAS Edition are directly or indirectly important for the prelims & main examination.

There are some topics which can be coded in answer writing of other topics in the main exam.

NEGOTIATORS IN SWITZERLAND, TRUMP WARNS IRAN

‘Can we turn over a new leaf?’: Vance, Ghalibaf lead talks

Iran links progress to Lebanon, US focus on Tehran n-plan



US Vice President JD Vance with President Donald Trump's envoys Steve Witkoff and Jared Kushner; (right) Iran Foreign Minister Seyyed Abbas Araghchi with Parliament Speaker Mohammed Bagher Ghalibaf at the Qatari-owned mountaintop resort of Buergenstock in Switzerland, Sunday. AP & REUTERS

Aamer Madhani, Seung Min Kim & Jamey Keaten
Obbuerger (Switzerland), June 21

US VICE President JD Vance said Sunday there was an opportunity to “turn over a new leaf” with Iran as the sides held talks aimed at building out the interim deal to end the war in Iran reached by the two sides last week.

Vance and US negotiators met with Iran's parliamentary speaker, Mohammad Bagher Ghalibaf, and Foreign Minister

E. EXPLAINED

Interim deal

The interim deal between the US and Iran is meant to stop fighting on all fronts, including Lebanon. It also gives negotiators 60 days to reach a nuclear agreement, but that can be extended.

Abbas Araghchi at a Swiss mountainside resort near Lake Lucerne.

Mediators from Pakistan and Qatar were also in the room for the direct engagement that, according to Iranian state media, lasted about 80 minutes. The US and Iranian negotiating teams also held separate private talks with Pakistani and Qatari officials.

The US is looking to get Iran locked into negotiations over its nuclear programme amid concerns it may be used for military purposes, which Iran denies. Vance also wants to push Tehran to commit to

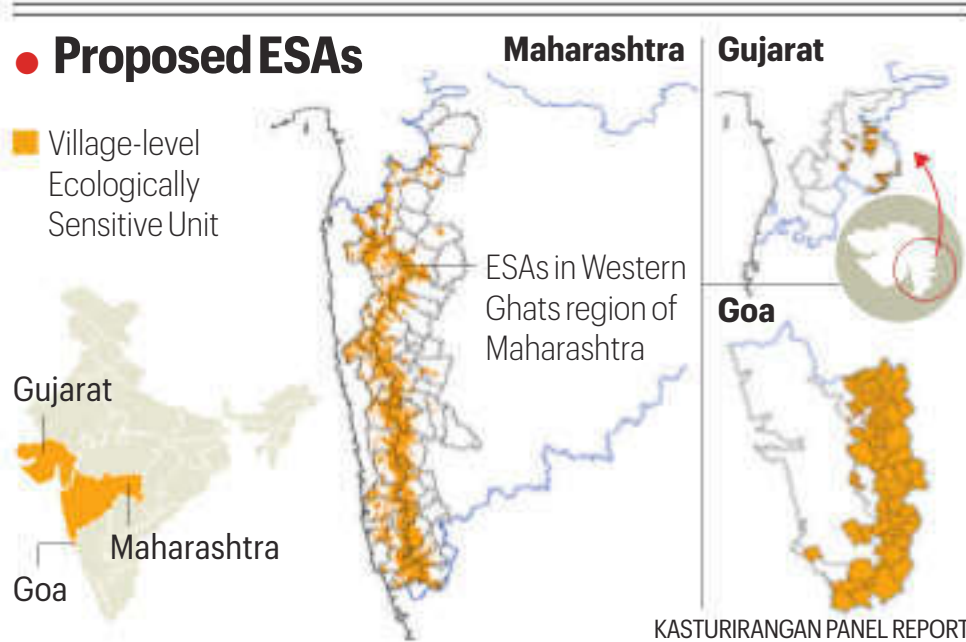
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Govt prepares to notify Western Ghats eco sensitive areas in at least three states

Nikhil Ghanekar & Amitabh Sinha
New Delhi, June 21

TWELVE YEARS after it issued the first draft notification, the Centre is finally ready to finalise and notify the demarcation of Ecologically Sensitive Areas (ESA) in the Western Ghats region, at least in the three states in which contentious issues have more or less been resolved, *The Indian Express* has learnt.

Over 56,000 square km of land in six states are proposed



to be demarcated as ESA, based on the 2013 recommendations of a high-level working group led by former ISRO chairman K Kasturirangan.

The Centre's original draft notification had come in 2014, and has undergone five revisions since then, but disagreements with the state governments on the areas earmarked for declaration as ESAs within their jurisdictions have not been fully resolved. Even now, the states of Kerala and

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- ASSOCIATED PRESS

Ghats

The Western Ghats are one of the eight 'hottest biodiversity hotspots' and the mountain chain was inscribed as a UNESCO world heritage site for being the exclusive home of hundreds of plants and animals.

Times Square to Nicobar: World marks Yoga Day



(Clockwise from above) People perform yoga at Times Square in New York; in Kozhikode, Kerala; Indian Coast Guard personnel in Andaman and Nicobar; and local residents in Srinagar, J&K, on the occasion of International Yoga Day on Sunday. PTI/SHUAIB MASOODI



PM’s message: Should aim to become more flexible at 40 than 20

Sweety Kumari
Kolkata, June 21

AS THE world marked the 12th International Yoga Day on Sunday, Prime Minister Narendra Modi said yoga has become the largest global community celebration, binding people across countries and cultures.

Kicking off the day’s celebrations at Kolkata’s iconic Red Road, Modi said June 21, the longest day of the year, or summer solstice, in the northern hemisphere, is now known to the world as a day of yoga.

“June 21, which marks the longest day on Earth, has now become the largest community celebration day because of yoga. Yoga brings people together. I congratulate the people of the world on this occasion,” he said.

He also emphasised this year’s theme for the celebrations, ‘Yoga for Healthy Ageing’, and said, “Yoga is not just physical exercise, not restricted to any age group; it is an expression of human spirit.”

He said the aim should be to become more flexible at 40 than at 20, and more energetic at 50 than at 30. He added that this ancient practice plays a crucial role in promoting physical health, mental well-being, and



PM Narendra Modi at Red Road in Kolkata, after kicking off the celebrations of 12th International Yoga Day, on Sunday. EXPRESS

active ageing, thereby improving the quality of life.

Flanked on the stage by Chief Minister Suwendu Adhikari and Governor R N Ravi, PM Modi practiced a series of asanas before walking through the crowds to interact with participants, occasionally pausing to correct their postures. He also invoked the philosophies of Rabindranath Tagore, Swami Vivekananda, and Rishi Aurobindo to underscore the spiritual and social depth of the practice. “Tagore believed that human identity lies in staying connected with the people around us. Yoga speaks of this connection. Maharishi Aurobindo used to say that our entire life is yoga. There-

fore, yoga is not just physical postures... it is the expression of consciousness in human life.”

Adhikari thanked Modi for elevating yoga to a global platform. Given the state’s political landscape, Adhikari claimed that although International Yoga Day began 12 years ago, it had never been officially celebrated as a government-backed initiative in West Bengal until the BJP government took office. The CM credited Modi for the achievement of “bringing yoga home”, while stressing that the tradition of yoga had been practised in Bengal for over a century.

The fervour was mirrored across the state and other parts of the country.

Monsoon revival likely this week, very heavy rainfall in Northeast

Heatwave in UP, MP, Maharashtra; ‘severe’ hot spell in Vidarbha until June 24, says IMD

Express News Service
Bengaluru, June 21

AFTER OVER a week of no movement, the southwest monsoon is likely to advance along the west coast and eastern India regions around the middle of this week.

The India Meteorological Department (IMD) on Sunday said, “Conditions are favourable for further advance of southwest monsoon into some more parts of Maharashtra, Telangana, Odisha, Jharkhand, Bihar and Chhattisgarh around June 23.”

The Arabian Sea branch of the monsoon had last progressed along south Konkan regions on June 8. The monsoon onset stands significantly delayed over Maharashtra, where it would have, under normal climatological circumstances,



A flooded Mumbai road after a brief spell of rain, Sunday. AKASH PATIL

covered the state by June 15. The delay has resulted in heatwave conditions continuing over the Vidarbha region.

On Saturday, Brahmapuri in Maharashtra and Banda in Uttar Pradesh remained the hottest locations in the country, with the maximum temperature touching 43.2 degrees Celsius. On Sunday, heatwave conditions continued over Vidarbha, where the day temperatures remained 5 to 7 degrees above normal.

Ahead of the next leg of the monsoon advancement, the

Met Department has warned of “very heavy” to “extremely” heavy rainfall (110 to over 200mm in 24 hours) along the northeast India region, which will continue till June 25. Parts of sub-Himalayan West Bengal and Sikkim will receive intense rain until Monday.

The IMD has said heatwave conditions would prevail over Chhattisgarh, eastern Madhya Pradesh until June 25, over eastern Uttar Pradesh till June 24. Vidarbha will experience “severe” heatwave until June 24, it has warned.

Yellow alert for Mumbai; water woes to continue

Pratip Acharya
Mumbai, June 21

MUMBAI EXPERIENCED light rain Sunday morning, with several areas in the island city and eastern suburbs recording brief showers between 6 am and 7 am. The IMD issued a yellow alert on Sunday, which is in effect until June 24. This alert indicates the possibility of moderate rain, accompanied by lightning, thunderstorms, and gusty winds up to 40 kmph. A yellow alert has also been issued for Thane, Palghar, Raigad and Sindhudurg.

Meanwhile, officials have stated that the rain will not affect the catchment areas of the seven lakes — Tulsi, Vihar, Tansa, Bhatsa, Modak Sagar, Upper Vaitarna and Middle Vaitarna — that supply water to Mumbai. “...Since these lakes are located deep in the hinterland, chances of their catchment areas receiving substantial rain do not seem favourable at this time,” Sushma Nair, scientist at IMD Mumbai, said.

Rakhigarhi skeletons to undergo DNA tests, facial reconstruction

Sukhbir Siwach
Chandigarh, June 21

THE ARCHAEOLOGICAL Survey of India (ASI) has sent nearly 5,000-year-old skeletons unearthed at Haryana’s Rakhigarhi, the largest known site of the Harappan civilisation, for scientific examination and facial reconstruction.

Four skeletons recovered during the latest excavation were sent to the Anthropological Survey of India (AnSI) in Kolkata on Friday, while DNA samples from one of the skeletons have been forwarded to the Birbal Sahni Institute of Palaeosciences (BSIP), Lucknow. Speaking to *The Indian Express*, Manoj Saxena, Superintending Archaeologist, ASI, said three skeletons, along with soil samples, had been sent to Kolkata, while only skeletal remains were sent in the fourth case. A fifth skeleton has been retained by the ASI.

Of the five skeletons recovered, three are believed to be female and two male. Preliminary assessments suggest all five individuals were between 30 and 40 years old,



The nearly 5,000-year-old skeletons were unearthed in Haryana’s Rakhigarhi, the largest known site of the Harappan civilisation. EXPRESS

though their exact ages will be determined through detailed scientific analysis.

Saxena described the discovery as significant as the remains date back nearly 5,000 years to the Harappan period. He said the studies would help determine whether the individuals died of natural causes or disease and, if disease was involved, identify the nature of the ailments.

The research is also expected to reveal details about their ancestry, height, physical

features and social status.

One of the key objectives of the study is facial reconstruction, which could provide a visual representation of what inhabitants of the Harappan civilisation may have looked like. According to Saxena, anthropological and DNA studies will be combined to produce a comprehensive report.

Researchers are also expected to reconstruct the palaeo-environment of the period to better understand the human populations that inhabited the region and identify areas requiring further study. Apart from the five complete skeletons, archaeologists recovered three additional burials from Mound No. 7, the ancient cemetery area of the site. However, these burials contained only fragmentary remains. One yielded only a skull, another contained damaged bones, while a third had a broken skull.

Archaeologists believe these remains were disturbed by agricultural activity over the years, as they were located just 15 to 20 centimetres below the surface.

• TECH

Guardrails in AI growth to protect developing nations

Global regulation can help with technology innovation while ensuring that the benefits of AI are widespread and equitable


EXPERT EXPLAINS
B RAVINDRAN

HEAD, CENTRE FOR RESPONSIBLE AI,
IIT MADRAS

THE TRANSFORMATIVE potential of artificial intelligence (AI) is becoming increasingly evident, and its impacts are already being felt, touching everything from daily life to broader industries. Consequently, efforts have begun to create global institutions and frameworks to regulate the development and deployment of AI systems.

Some preliminary institutional structures have already been set up. Last year, the United Nations General Assembly established a Global Dialogue on AI in which every country has been invited to participate. An Independent International Scientific Panel on AI is tasked with making periodic scientific assessments to enrich the Global Dialogue discussions.

Professor B Ravindran, head of the Centre for Responsible AI at IIT Madras, is the only Indian among the 40 people appointed to the Panel for a three-year term. He spoke to **Amitabh Sinha** about the need for a system of global AI governance, why it matters for developing countries like India, and what a commons system for AI tools and resources could look like.

What is the International Scientific Panel on AI supposed to do?

This is the first time a global scientific body on AI has been set up. It is supposed to take a periodic view of the state of science around AI and inform the Global Dialogue on AI.

Our primary responsibility would be to produce reports on the scientific aspects of AI. This body would not get into the politics or policy side of AI discussions. That is the responsibility of the Global Dialogue, in which governments would participate.

Why is a global governance structure on AI needed? Are we moving in that direction?


Visitors at the AI Impact Summit in New Delhi this year. PRAVEEN KHANNA

• Where AI regulation stands

• Several parallel attempts are being made to regulate the development and deployment of artificial intelligence. Most of these structures have voluntary participation, varying degrees of legal force, and focus on specific aspects like standards, safety and ethics.

• The most comprehensive legal framework for AI is the European Union AI Act of 2024. EU prioritised safe, transparent, non-discriminatory and environmentally friendly AI systems.

• Annual global AI summits, whose most recent edition was held in New Delhi in February, have also resulted in mechanisms that could feed into the development of global governance models on AI.

Would international regulation also control the risks from AI?

There are definite risks, no doubt. People talk about AI's capabilities in building the next generation of biological weapons or the next most powerful chemical weapons. We have international agreements and treaties that control biological or chemical weapons. Now, people might also want to bring about international frameworks that control AI models and the tools that enable these weapons to be built.

While that is logical, the fear and worry are that it might take the shape of non-proliferation language. There might be attempts to propose that only certain countries or companies can develop safe and responsible AI, and therefore, they should be the only ones allowed unrestricted development and use of AI systems. That would lead to something similar to the global nuclear regime.

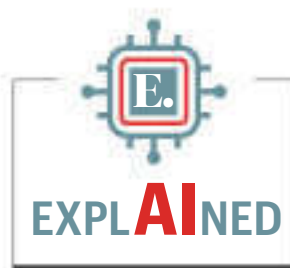
I am worried about the dialogue on AI moving in that direction. Hopefully, it will not happen, but I think we have to be on constant vigil against it.

You were personally involved in developing the framework for Trusted AI Commons, which was one of the main outcomes of the New Delhi AI Impact Summit in February. What is that, and how would it help?

It can be seen as another part of the global governance structure. The Trusted AI Commons is a repository of various tools needed to develop and deploy AI systems in a safe and responsible manner.

Suppose someone wants to test an AI system meant for agriculture. Trusted AI Commons would be like a one-stop shop to figure out what tools are available to test AI deployment, what the benchmarks, datasets, and even protocols are. Several resources will be housed in the Trusted AI Commons.

To begin with, this Commons will be hosted and managed by India. The India AI mission has to identify a suitable vehicle to do that. Trusted AI Commons will not commission building these tools right away, but many organisations are already working on it. This includes the Centre for Responsible AI at IIT Madras. Companies like Google have also created their own tools. It is called Commons because it will be open, accessible, and with very liberal licensing requirements.



A weekly column decoding AI's new frontiers, how they shape the world and our place in it

Safety and access

AI Commons is being built as a repository of accessible tools, datasets and other resources to test AI systems.

be within national boundaries, it can lead to significant concentration of power. Already, a few countries have the infrastructure and resources to support the entire AI ecosystem.

The other perspective is of deployment. Many countries in Asia or Africa might not have the resources or capacities to frame robust regulations to protect their interests. There would always be a danger of them being reduced to digital colonies.

In this context, having a globally agreed-upon set of minimum regulations would be useful.

AI is a very transformative technology, with a capability of the order of the steam engine. It would be awful if, owing to a lack of regulatory capacities, certain developing countries were denied the benefits of AI. It will be a disservice to humanity at large.

This is why international agreements and arrangements are important. We need to ensure that the interests of countries that are not AI developers themselves are also protected, and they continue to benefit from advancements in this technology.

• DEFENCE

Three distinct strategic roles of the three ships inducted into Navy

Sushant Kulkarni
Pune, June 21

THE NAVY inducted three ships on Sunday — one that can fight far out at sea, one that can map the sea, and one that can hunt submarines close to the coast.

The three vessels — the Brahmos-armed stealth frigate INS Dunagiri, the deep-water survey vessel INS Sanshodhak, and the Anti-Submarine Warfare Shallow Water Craft (ASW SWC) INS Agray — have been built by Garden Reach Shipbuilders & Engineers (GRSE) in Kolkata. According to the Ministry of Defence, the three together have more than 75 per cent indigenous content, and involve over 200 MSMEs.

What are these three ships?

INS DUNAGIRI: The largest and most heavily armed of the three is INS Dunagiri. It is a frigate, smaller than a destroyer but large enough to operate far from the coast.

It is part of the Navy's Project 17A, under which a new generation of stealth guided-missile frigates are being built in India. Dunagiri carries weapons such as BrahMos surface-to-surface missiles, sensors including Multi-Function Surveillance, Track



PM Modi presided over the commissioning ceremony in Kolkata on Sunday. PTI

And Guidance Radar (MFSTAR), electronic warfare systems and anti-submarine weapons. "At the strategic level, these multi-mission frigates are capable of operating in a 'blue water' environment — deep sea far from shore — dealing with both conventional and non-conventional threats," a serving Navy officer said. Other vessels in the class are INS Nigiri, Himgiri, Taragiri, Udaygiri and Vindhayagiri.

INS SANSHODHAK: This is a Survey Vessel

— Large (SVL). Its job is to measure and map the sea: the depth of waters, seabed features, approach channels to ports, navigational routes, and oceanographic data. It is equipped with systems such as autonomous underwater vehicles, remotely operated vehicles and multi-beam echo sounders. These help the ship collect data from the surface and underwater.

"Warships and submarines do not operate in an empty blue space. They move through waters shaped by depth, seabed features, currents, reefs, channels, ports, underwater slopes and coastal clutter. Knowing this environment helps ships move safely, submarines plan routes, ports update charts, and forces prepare for operations. It also supports civilian needs such as safe shipping, disaster relief, ocean research and coastal development," said a Navy officer who has served on survey ships.

INS AGRAY: The smallest vessel of the three, it has a highly specialised role. Agray belongs to the Arnala-class of smaller warships built specifically to detect and attack submarines in shallow waters near the coast, ports, naval bases and important sea approaches. Agray carries lightweight torpedoes, indigenous anti-submarine rocket

launchers and sonar systems.

"Coastal waters are difficult places to find submarines. They are noisy, busy and cluttered. Fishing boats, merchant ships, seabed features and coastal activity can make submarine detection harder than in the open ocean. That is where ASW-SWCs such as Agray come in," said an officer.

What is the larger message behind commissioning them together?

The importance of Sunday's ceremony is that the three ships add three different capabilities at once.

As the Indian Ocean becomes more contested, the triple-commissioning "is a sign of how the Navy is building capability in layers: big warships for distant operations, survey ships to understand the sea, and smaller submarine-hunters to guard coastal waters," said an officer.

It is also an industrial signal. That three different kinds of ships, with different missions and technologies, can be built domestically and commissioned together shows the growing maturity of India's naval shipbuilding ecosystem.

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The Editorial Page

MONDAY, JUNE 22, 2026

Road to energy security passes through our farms


FROM PLATE TO PLOUGH

 BY ASHOK GULATI
AND SUBHODEEP BASU

THE WEST Asia crisis has exposed a structural vulnerability in India's growth story: Its heavy dependence on imported fossil fuels. Whether it is power generation, transport or fertiliser production, India's energy architecture remains deeply import-dependent. For a fast-growing economy, this is not merely a macroeconomic concern; it is a strategic risk.

But energy insecurity is not India's only challenge. Climate change is increasingly threatening agricultural livelihoods. The forecast of a strong El Niño in 2026 does not augur well for farm incomes. The question, therefore, is not only how India secures its energy future, but also how it builds income resilience for its farmers, who comprise 43 per cent of the workforce. The answer may lie in turning India's *annadata* into *urjadata*, from producers of food to producers of energy, too.

Prime Minister Narendra Modi's push towards renewable energy is commendable. As of March 2026, renewable energy capacity stood at about 274 GW, with solar alone crossing 150 GW. But the model so far adopted in solar energy is mainly large solar parks and PM Surya Ghar. The PM-KUSUM scheme is basically for solar pump sets. This is not sufficient for agricultural transformation. We

lay out below a case for agriphotovoltaics on farmers' fields that has potential not only to generate clean energy in rural areas but also to halve the government's power subsidy, while also increasing farmers' incomes manifold. If promoted by PM Modi on priority basis, it can transform not just agriculture but the very growth story of India. Let us delve a little deeper into it.

Agriphotovoltaics (Agri-PV) integrate solar power generation and crop cultivation on the same piece of land. This allows farmers to harvest both sunlight and crops. Solar panels are mounted at a height of about 3.5 metres, which permits cultivation beneath them while generating electricity for sale to discoms. The result is dual use of land and dual streams of income.

Unlike agricultural income, which is vulnerable to weather shocks, pest attacks and market fluctuations, solar income is stable. In effect, solar becomes a "third crop" that provides a dependable source of earnings when conventional crops fail.

ICRIER, with support from Kotak Mahindra Bank's CSR initiative, has created a pilot 600 KW solar plant in Rajasthan. The State Bank of India provided a loan of Rs 1.4 crore, and the farmer himself put in Rs 60 lakh. The project has increased the farmer's income from about Rs 40,000 per acre from wheat and bajra cultivation to nearly Rs 4 lakh per acre through a combination of energy sales and shade-tolerant horticulture, demonstrating a tenfold income enhancement potential.

The significance of Agri-PV extends beyond farmers' income. It also addresses a longstanding distortion in India's power sector. Agriculture consumes nearly 2,60,000 GWh of electricity annually, yet pays tariffs far below the actual cost

of supply. According to the Comptroller and Auditor General of India's recent report, Steering India's Power Sector Towards Viksit Bharat (2026), the average cost of supplying electricity in India is around Rs 8.5/kWh, and assumed revenue realisation from agricultural consumers close to Re 1/kWh. This implies an effective subsidy of at least roughly Rs 7.5/kWh. India's annual power tariff subsidy bill is approximately Rs 2.35 lakh crore, with agriculture accounting for roughly 85 per cent. The true cost of power subsidy to agriculture may be even higher as the cost of supply to agriculture is higher than Rs 8.5/kWh, once technical losses, long feeder lines and the costs of maintaining rural distribution infrastructure are taken into account. This would take the farm power subsidy to over 90 per cent of total tariff subsidy. No wonder that, despite repeated reforms and financial bailouts, discom finances remain under severe stress.

Agri-PV offers an opportunity to fundamentally alter this equation. By generating electricity at the farm level and feeding it into local distribution networks, it reduces transmission losses and infrastructure costs while transforming farmers from consumers of highly subsidised power into producers of clean energy. However, scaling Agri-PV requires supportive policy. Elevated structures make Agri-PV systems 15-20 per cent

more expensive than conventional solar projects, creating a significant entry barrier for farmers.

A differentiated feed-in tariff of say Rs 4.5/kWh, coupled with targeted support under PM-KUSUM, can significantly improve project viability and accelerate adoption. From the perspective of discoms and state governments, such support should be viewed not as an additional subsidy but as a smarter

reallocation of existing expenditure. Even at a feed-in tariff of Rs 4.5/kWh, discoms would be procuring clean power at roughly half the effective cost of supplying power to agriculture. In effect, every unit of electricity generated through farmer-owned Agri-PV has the potential to reduce the subsidy burden by nearly half while simultaneously creating a new source of farm income.

The policy implications are clear. Existing support mechanisms under PM-KUSUM Components B and C already provide capital subsidies for standalone solar pumps and the solarisation of grid-connected agricultural pumps. If public support is considered justified for solarising irrigation, a strong case can be made for extending similar support to farmer-owned Agri-PV projects under Component A that simultaneously generate clean energy and preserve farm incomes. Targeted capital subsidies and an attractive feed-in-tariff can be game-changers, and a win-win for the government as well as farmers.

Institutional innovation will be equally important. Just as milk cooperatives transformed India's dairy sector, solar cooperatives can aggregate small-holders, improve access to finance, strengthen bargaining power and manage power sales collectively. Such cooperative institutions can become the vehicle through which the Agri-PV revolution can be unleashed.

If PM Modi takes it up on priority, he can not only transform India but also set an example for the Global South, especially Africa, under the theme of "One Sun, One Earth, and One Grid". That would bring relief from fossil fuels while augmenting farmers' incomes. Can he do that? Only time will tell.

Gulati is distinguished professor and Basu is research fellow at ICRIER. Views are personal

On AI, US & China have an outdated arms race mindset


BHARATH REDDY

OVER THE past few weeks, two governments showed us how they are thinking about AI. Beijing restricted overseas travel for top AI engineers from private firms like DeepSeek and Alibaba. Washington issued a directive to Anthropic to suspend its frontier models for foreign nationals, including its own employees. A commercial product used by millions of people was pulled off the shelf. One country treated software engineers like nuclear physicists, and the other treated a software product like a munition. Both policy actions flow from the same story about what AI is.

National security thinking in the US and China treats AI as a zero-sum arms race or a weapon of mass destruction to be contained through non-proliferation. This narrative has been promoted by frontier labs to gain market share and shape the regulatory landscape. Other policy instruments seen recently, such as export controls and hardware tracking of chips, also stem from this narrative. The framing makes coercion feel like prudence.

However, this analogy misrepresents the technology. Unlike rare, controllable fissile material, AI models are mathematical artefacts that are easily reproducible and constantly diffusing. Chinese firms built competitive models despite chip export controls, and powerful open-source models remain freely downloadable. The bottlenecks to unlocking productivity lie in widespread adoption: Bridging the capability-reliability gap, learning curves, organisational reforms and regulatory compliance.

A more honest framing is the geopolitical innovation race. Here, a dense web of public and private actors across the world compete and collaborate, driven by a mix of national security, economic advantage, and market incentives. The policy instruments that follow are constructive: Compute investment, skilling, industrial policy, strategic partnerships, organisational reforms.

The response for a middle power like India is to lean into AI as a general-purpose technology while acknowledging the geopolitical innovation race. The value lies not just in inventing the frontier but also in diffusing it widely and well. The Supreme Court's draft guidelines on the use of AI tools in courts are a telling example. The returns of the AI era will accrue to the countries that put it to work in health, agriculture, law, and public services.

That points to three priorities. Diffusion over denial: Spend political energy on adoption infrastructure, regulatory clarity, and sector deployment. Plurilateral resilience over self-sufficiency: Coalitions of the like-minded — India and the EU have complementary strengths — can share open-source development costs and build trusted supply chains. Finally, open over closed: The Fable 5 and Mythos 5 suspension shows that dependence on a nationally controlled, closed model risks access being withdrawn overnight. Open-weight models, open standards like RISC-V, and open alternatives to proprietary stacks are resilient because no single government can switch them off.

The writer is associate fellow, Takshashila Institution

The response for a middle power like India is to lean into AI as a general-purpose technology while acknowledging the geopolitical innovation race

India has spelt red lines to US. Now seal trade deal

THE LAST year-and-a-half has seen strains in the India-US relationship over trade, Russian oil, and claims of American mediation during Operation Sindoor. Despite high-level engagement, including US Secretary of State Marco Rubio's recent visit to India, restoring trust between the two sides remains a work in progress. India is not the only country whose diplomacy has been challenged dealing with the mercurial US President, Donald Trump. Italian PM Giorgia Meloni's once-warm relationship with Trump has soured over Rome's reluctance to support Washington in its war with Tehran. Ukraine's Volodymyr Zelenskyy and Canada's Mark Carney are no strangers to this predicament. Trump's maverick style of statecraft has forced allies and rivals alike to recalibrate their approach to Washington. It is in this context that the meeting between Trump and Prime Minister Narendra Modi — their first in 16 months, on the sidelines of the G7 summit last week — assumes significance. The Prime Minister welcomed the progress made in peace efforts in West Asia and underlined that "mutual trust is the most important strategic asset today".

The US President, through his military actions, weaponisation of tariffs and transactional diplomacy, has eroded the post-war international order, and India has not been immune to the consequences. The question is whether New Delhi can quickly come to terms with the structural shifts underway in American foreign policy. Given that the US is India's largest economic partner, not just in trade, but also in investment, technology and higher education, the fundamentals of the relationship remain strong. New Delhi has no choice but to adapt and manage Trump for the remaining two-and-a-half years of his term; neither open confrontation nor submissive behaviour is likely to work. Prime Minister Modi's approach at the G7 — crediting Trump for his peace efforts while simultaneously working towards a trade deal to limit tariffs, as US Trade Representative Jamieson Greer visits India this week — appears to be the most effective course available.

That does not mean India should hesitate to defend its interests. PM Modi raised with Trump the safety of Indian seafarers after External Affairs Minister S Jaishankar lodged a protest with Rubio over US attacks in the Gulf that killed three Indian mariners earlier this month. The task of diplomacy is to make India's red lines clear while navigating crises beyond its control. What remains in India's control is strengthening its leverage by modernising the economy, defence capabilities and technology ecosystem. Achieving those goals will require deeper engagement with the West, particularly the US. India's challenge is to preserve both partnership and autonomy.

When doctors can move, healthcare benefits

IN A first-of-its-kind reform, Andhra Pradesh has thrown open its doors to doctors from across India by removing state registration hurdles. An MBBS graduate holding a recognised medical qualification and valid registration with any state or UT medical council in the country will now be allowed to practise in the state, without a separate Andhra Pradesh Medical Council registration or a No Objection Certificate (NOC) from the council where they are registered. The order cuts through a bureaucratic roadblock that has long deterred doctors — particularly young graduates and specialists — from moving to states where there are better opportunities. With Andhra Pradesh's registered doctor base standing at roughly 1.05 lakh for a population of nearly 5 crore, and the country's overall doctor-population ratio hovering at 1:811 against the WHO benchmark of 1:1,000, this is likely to help address shortages in specialist care and strengthen the state's ambition to attract medical talent from across India.

The significance of the decision is not limited to healthcare administration alone. In a country that routinely speaks of the ease of living and doing business, domicile preferences in public employment and local registration requirements have often tested the spirit, if not always the letter, of constitutional guarantees. Driven by political and administrative considerations, states frequently create procedural hurdles that limit opportunities. For instance, in an attempt to stem high attrition rates, Bihar recently made it mandatory for government doctors to complete three years of continuous service before becoming eligible for NOCs. There are also other tangible costs for those seeking to relocate. Re-registration fees at state councils typically range from Rs 5,000 to Rs 10,000, while obtaining an NOC from a home-state council — a prerequisite for re-registration elsewhere — can take weeks and involve additional paperwork.

Andhra Pradesh's move also aligns with the broader objective behind the National Medical Commission's efforts to create a seamless national registration framework. The "One Nation, One Registration" platform was launched in August 2024 to eliminate impediments to professional mobility. Yet only around 1,800 registration certificates have been issued so far, while more than 30,000 applications remain pending verification at state medical councils. By lowering barriers to mobility, Andhra Pradesh has not only moved to strengthen healthcare delivery, it has shown that states need not wait for administrative bottlenecks to be resolved before adopting the principle underlying the reform.

The Ideas Page

MONDAY, JUNE 22, 2026

In time of distrust, Constitution's promise of fraternity matters more than ever



S Y QURAISHI

EVERY SCHOOLCHILD can recite the Preamble to the Constitution: Justice, liberty, equality and fraternity. The first three have generated debate in courts, politics and academia. The fourth, however, has passed almost unnoticed. Yet fraternity may be the most consequential of all the constitutional values because it is the one value that law alone cannot create.

The neglect is curious because the Constitution does not treat fraternity as an afterthought. It appears in the same sentence as the other three, carrying equal constitutional weight. The framers placed it there deliberately. The question is why we have spent 75 years largely ignoring it, and what that neglect has cost us.

The answer begins with a distinction that was explicit during the Constituent Assembly debates but has since faded from constitutional discourse. Liberty and equality define the relationship between the citizen and the state. Fraternity defines the relationship between citizens themselves. That distinction makes all the difference. If the state violates liberty or equality, courts can intervene. But what recourse exists when citizens cease to regard one another as fellow citizens? When disagreement hardens into hostility and difference into suspicion, fraternity begins to disappear. B R Ambedkar understood this with unusual clarity. In his final address to the Constituent Assembly on November 25, 1949, he warned, "Without fraternity, equality and liberty will be no deeper than coats of paint." His warning stemmed from a frank assessment of Indian society. "How can people divided into several thousands of castes be a nation?" he asked. He was reminding us that nationhood is not a given but an achievement, and fraternity both its condition and proof. The Constitution's concern with fraternity did not end with the Preamble. Article 51A(e) makes it a Fundamental Duty of every citizen to pro-

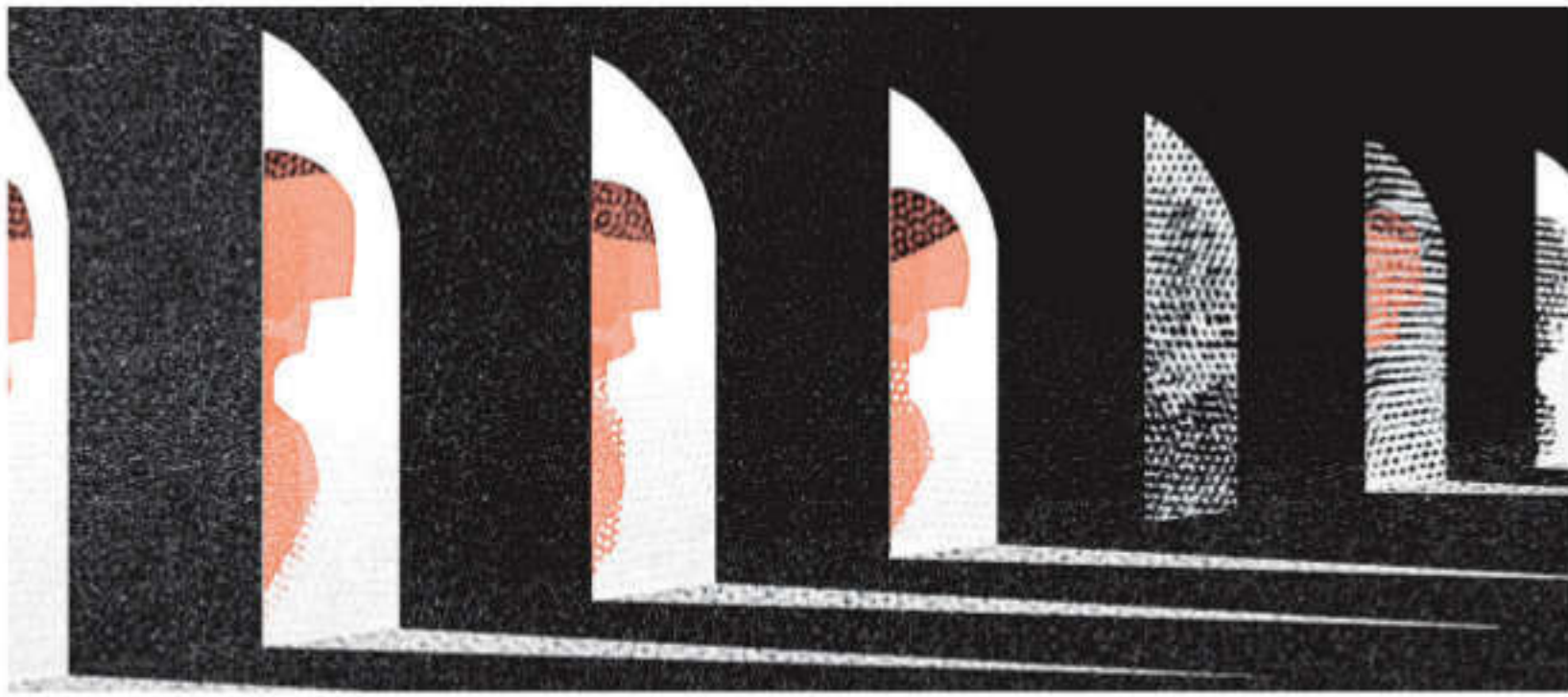


ILLUSTRATION: C R SASIKUMAR

mote harmony and the spirit of common brotherhood across religious, linguistic and regional differences.

Seventy-five years later, the original anxieties have not diminished. Though we have never been more connected technologically, we have rarely been more estranged socially. Public debate has grown louder, sharper and more vicious, while mutual understanding has become thinner. The greatest damage caused by hate-filled discourse — whether political, religious or digital — is not that it offends. Democracies have always survived offence. Its deeper damage lies in the erosion of trust.

Trust is the invisible infrastructure of a functioning society — the assumption that strangers can deal with one another in good faith. When that assumption erodes, no law can fully restore it. The economy depends upon it. Democracy depends upon it.

Liberty and equality are justiciable, enforceable and actionable before a judge. Fraternity belongs to a different category. If liberty and equality are the fundamental rights of citizens, fraternity, for me, is the fundamental right of the nation. No court can manufacture mutual respect. No statute can create a sense of common belonging. No government programme can legislate trust. Yet it would be wrong to conclude that fraternity lies entirely beyond the reach of law and the government and the courts are helpless spectators.

No court can manufacture mutual respect. No statute can create a sense of common belonging. Yet it would be wrong to conclude that fraternity lies entirely beyond the reach of law

The Constitution and the criminal law contain a substantial architecture designed to protect social harmony. Article 14 guarantees equality before the law. Articles 15 and 25 protect citizens against discrimination and safeguard freedom of faith. Article 51A(e) imposes a duty to promote harmony and common brotherhood. Criminal law goes further. Sections 153A, 153B, 295A and 505 of the IPC, along with provisions protecting places of worship, were enacted to preserve civic peace and prevent communal hostility.

These provisions may not create fraternity, but they exist to prevent its systematic destruction. The challenge, therefore, is not the absence of legal safeguards. It is their uneven enforcement. Courts cannot command affection, but they can act against hatred, intimidation and attacks on equal citizenship. The law cannot manufacture fraternity, but it can — and must — prevent its destruction.

It is worth asking what 75 years of neglecting fraternity has produced. Consider what the government's own data tell us. For more than a decade, roughly 2 lakh Indians have been surrendering their citizenship every year, amounting to nearly 9 lakh in the last five years alone, according to information placed before Parliament. Many of those leaving are successful entrepreneurs and professionals. Their decisions reflect multiple factors, but the scale of the phenomenon raises a legitimate question: What role do social climate and civic

confidence play in such choices?

History suggests that prolonged social conflict leaves economic scars that persist long after the headlines disappear. India has already learned this lesson. Punjab was once the most prosperous state in the country, with the highest per-capita income as recently as the mid-1980s. The decade-long militancy and communal discord that followed devastated the investment climate. Punjab slipped from first place in per-capita income rankings and, between 2014-15 and 2022-23, it ranked 18th in economic growth among 21 major states. The damage was not merely economic; poisoned atmospheres take generations to clear.

The irony is difficult to miss. We have spoken of India becoming the world's third-largest economy and a fully developed Viksit Bharat by 2047. Yet public institutions have often appeared insufficiently concerned about an atmosphere of anxiety and periodic lawlessness that seems to corrode such ambitions. Lynchings, targeted demolitions and the normalisation of communal hostility are not merely law-and-order failures. They are fraternity failures. When such acts go unpunished, they send a signal — not just to minorities, but to every entrepreneur, investor and family weighing its future.

Polarisation may have delivered electoral dividends to those who have practised it. But a nation cannot continuously inflame its divisions for short-term political advantage and expect to contain its economic consequences.

Fraternity is not a pious wish inscribed in a preamble and ceremonially quoted on national occasions. It is the social oxygen without which neither liberty nor equality can survive, and without which no economy, however ambitious its targets, can truly flourish. Ambedkar warned that without fraternity, liberty and equality would be no deeper than coats of paint. For three-quarters of a century, we barely noticed the paint peeling. We are beginning to see the cracks now.

The framers placed all four values together for a reason. It may be time we did the same.

Quraishi is former chief election commissioner of India and author of An Undocumented Wonder: The Making of the Great Indian Election

Ease of justice has brought legal system closer to people



ARJUN RAM MEGHWAL

JUSTICE HAS always been a vital and integral pillar of human civilisation. Judicial institutions have served as the bridge that enables every stakeholder to develop a sense of trust, belonging, and association with the larger pursuit of justice and collective good. This necessitates the creation of a robust justice ecosystem — one that not only ensures access to justice but also facilitates ease of living for every citizen.

Under the leadership of Prime Minister Narendra Modi, the governance paradigm has facilitated ease of living through a bottom-up approach while implementing a nation-first approach from the top down. Judicial reforms have encompassed legislative modernisation, institutional strengthening and digital innovation. The mantra 'ease of justice' incorporates ease of engagement for litigants, ease of working for advocates and judges, and ease of understanding for citizens.

From the litigant's perspective, services like Tele-Law and Nyaya Bandhu to promote pro-bono work under the Designing Innovative Solutions for Holistic Access to Justice scheme have made justice delivery accessible and affordable. More than 11.2 million beneficiaries from rural and remote areas have benefited from free pre-litigation legal consultations under Tele-Law. E-filing services and e-Seva Kendras have eased the litigant's interactions with the judicial ecosystem.

For advocates and judges, a push to upgrade both digital and physical infrastructure has enhanced the ease of working. As the subordinate judiciary serves as the first point of contact for the vast majority, strengthening it is an imperative. The number of court halls has increased from 15,818 in 2014 to 22,712, supported by the disbursement of Rs 9,400.40 crore since 2014 for the development of state-of-the-art integrated court complexes. Further, the e-Courts Phase-III project seeks to transform courts into end-to-end digital, paperless, and AI-enabled institutions of justice delivery. Initiatives like videoconferencing, virtual courts, and livestreaming court proceedings have brought the judiciary closer to the people.



Ease of understanding for citizens is an essential pillar of the broader ease of justice framework, particularly in a linguistically diverse country like India. AI-powered natural language processing tools such as the Supreme Court Vidhik Anuvaad Software and Bhashini are translating SC judgments and orders into 18 Indian languages. The National Judicial Data Grid provides single-click access to the analysis of information relating to over 340 million court orders.

The new criminal laws have incorporated contemporary realities. The integration of e-courts, e-prosecution, e-prisons, and e-forensics with the Crime and Criminal Tracking Network and Systems is a transformative step. The Nyaya Shruti platform is facilitating virtual appearances and the recording of witness testimonies so efficiently that when a court grants bail to a citizen, the digital bail order immediately reaches the prison portal, eliminating the delays that traditionally hindered timely release.

The sanctioned strength of high court judges has increased from 906 in 2014 to 1122; that of the Supreme Court was increased from 31 to 34 in 2019, and has now been raised to 38. The appointment of 1,175 high court judges and 77 Supreme Court judges from diverse backgrounds in the past 12 years reflects the executive's commitment to strengthening the judicial ecosystem.

Reducing more than 40,000 compliances and repealing 1,725 redundant and obsolete colonial-era laws has contributed to the ease of doing business. Strengthening arbitration law, setting up institutions like the India International Arbitration Centre, and promoting mediation through the dedicated Mediation Act, 2023 reflect India's global leadership in alternative dispute resolution. India's leadership ensured the adoption of the Gandhinagar Declaration, a collective commitment to strengthening mediation and arbitration, at the BRICS justice ministers' meeting in 2026.

As India advances towards the goal of Viksit Bharat by 2047, we remain steadfast in our commitment to building a future-ready justice system — one that is resilient, innovative, inclusive, and inspired by the collective aspirations of 140 crore Indians.

The writer is Minister of State for Law and Justice (I/C) and Minister of State for Parliamentary Affairs, Government of India



ASHOK KHEMKA

Electoral wrongs can be corrected before it's too late

THE CONTROVERSY over the rejection of a nomination paper in the recent Rajya Sabha election from Madhya Pradesh raises an important constitutional question: How should the legal system respond when a returning officer (RO) is alleged to have unlawfully excluded a candidate from an election, yet the constitutional commitment to uninterrupted elections restricts judicial intervention until the electoral process is over?

The issue highlights a tension in election law. The Constitution seeks to ensure that elections proceed without disruption and are conducted in accordance with law. These objectives are complementary in theory but can occasionally come into conflict. The conventional constitutional response is found in Article 329(b), which generally bars judicial interference in electoral matters once the election process has begun. It channels disputes into an election petition after the declaration of results. This principle, upheld by the Supreme Court, serves an important democratic purpose. Elections cannot be subjected to litigation at every intermediate stage. Yet Article 329 was never intended to create a zone of immunity for electoral illegality.

An erroneous rejection of nomination affects not merely the rights of a candidate but the structure of the election. After polling, the consequences of such exclusion may become difficult to remedy in a meaningful way. The question is

Election petitions are often decided after substantial delays. A successful petition may establish a legal wrong but offer little practical means of correcting it

whether the constitutional framework provides a corrective mechanism.

The answer may lie in a fuller understanding of Article 324 — it vests in the Election Commission of India the "superintendence, direction and control" of elections. Judicial decisions have recognised that Article 324's supervisory authority enables the ECI to safeguard the integrity of the electoral process. Although ROs exercise statutory powers under the Representation of the People Act, they do so under the ECI's constitutional control. If the body has the authority to supervise electoral conduct, ensure compliance with election law and even countermand elections in exceptional circumstances, it is difficult to conclude that it lacks the power to correct a legal error by an RO in the scrutiny of nominations.

Such oversight need not assume the character of conventional appellate jurisdiction. Every dissatisfied candidate cannot be permitted to seek the reconsideration of routine factual determinations. However, a limited revisional jurisdiction to correct jurisdictional errors, manifest illegality, or clear misinterpretation of statutory provisions would enable serious legal disputes to be resolved promptly before polling and, more importantly, harmonise Articles 324 and 329.

The SC's cautious approach was justified. However, constitutional adjudication must remain attent-

ive to the efficacy of remedies. Election petitions are often decided after substantial delays. By the time a final judgment is rendered, elected representatives may have served a significant part of their term. Political alignments may have shifted. Members of the electoral college may have resigned, died or become disqualified. A successful petition may establish a legal wrong but offer little practical means of correcting it.

That's why Article 142 is relevant — it empowers the SC to pass such orders as may be necessary for doing complete justice in a matter before it. It is not a licence to disregard constitutional limitations, nor can it routinely displace statutory remedies. It exists for exceptional situations. Where a candidate alleges exclusion on a ground that is ex facie contrary to law, where the ECI has not exercised corrective supervision under Article 324, and where postponing scrutiny until an election petition may render effective relief impossible, a carefully tailored exercise of Article 142 may be constitutionally defensible. This would not amount to repudiating Article 329, but an attempt to reconcile the constitutional commitment to electoral continuity with the equally important commitment to meaningful justice.

The controversy reveals a structural gap. The challenge is not to choose between the three articles, but to harmonise them.

The writer retired as a senior IAS officer. Views are personal



RAJESH GOPAL

INDIA'S DEVELOPMENT trajectory is entering a decisive phase. Infrastructure expansion, renewable energy transitions and economic growth are accelerating, often in the landscapes that underpin the country's ecological security. Forested regions, many overlapping with tiger habitats and wildlife corridors, are increasingly at the centre of this transformation. This raises a difficult question: How much ecological cost is too much?

For decades, India has relied on regulatory instruments such as Environmental Impact Assessments (EIA), forest clearances and compensatory afforestation to manage this balance. More recently, the idea of Green

GDP — adjusting economic output to reflect environmental loss — has gained attention. Yet these approaches are often difficult to translate into clear decision thresholds. Policymakers need not merely more data but a simple, credible benchmark to recognise when economic activity begins to undermine its own ecological foundation. An answer may lie in a principle that governs all natural systems. In every ecosystem, energy flows through a hierarchy — from plants to herbivores to predators. At each step, most of the energy is lost, leaving only a small fraction available for the next level. This principle, described by Lindeman's 10 per cent Law, explains why nature forms pyramids.

In much of India's forests, the tiger is at the pyramid's apex. Its survival depends on a large and stable prey base, which in turn depends on healthy vegetation, water sys-

tems and minimal disturbance. Even small disruptions at the base can cascade upward, destabilising the system. This ecological reality offers an important economic insight: Systems can absorb only limited stress before they begin to degrade.

Recent estimates place the economic cost of environmental degradation in India at around 9-11 per cent of GDP. A precautionary approach would be to treat this as an ecological threshold

Recent estimates place the economic cost of environmental degradation in India at around 9-11 per cent of GDP. A precautionary approach would be to treat this as an ecological threshold

gical threshold — a point beyond which the economy is not operating sustainably and is drawing down its natural capital. Development must pay a minimum ecological due.

This idea acquires particular urgency in tiger landscapes. India's tiger reserves and corridors are embedded within landscapes that support agriculture, infrastructure and local livelihoods. They also provide critical ecosystem services, including water regulation, carbon storage and biodiversity conservation. These systems also operate under tight ecological margins. Incremental pressures — linear infrastructure, mining, unsustainable extraction and land-use change — can accumulate rapidly. The consequences are already visible in many regions: Rising human-wildlife conflict, fragmentation of corridors and stress on prey populations. In such contexts, a simple threshold can

serve as an early warning. If the cumulative ecological cost of activities within a landscape begins to exceed a notional 10 per cent of its total economic or ecological value, it should trigger a policy pause — not to halt development, but to reassess its scale, location and design. In more fragile wildlife corridors, an even more conservative threshold of 5-8 per cent may be appropriate. Such an approach can be incorporated into EIA processes, landscape-level planning in notified corridors, green accounting frameworks and infrastructure design.

In nature, as in economics, systems that consume their foundations do not collapse immediately. But they do, eventually — and often without warning.

The writer is secretary general, Global Tiger Forum

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IRAN SAYS LEBANON A FOCAL POINT AS NEGOTIATORS MEET IN SWITZERLAND FOR PEACE TALKS

Ceasefire holds, but Israel says free to act against threats in Lebanon

Defence Minister Katz says troops to stay in Lebanon security zone

Reuters

Beirut, Jerusalem, June 21

ISRAELI SOLDIERS are free to act without restriction to eliminate threats in Lebanon, Israeli Defence Minister Israel Katz said on Sunday, adding troops remained in position in what Israel refers to as a security zone.

A ceasefire with Iran-backed Hezbollah took effect on Friday after months of escalating violence, but on Saturday Israeli strikes killed at least 20 people in Lebanon, Lebanon's state news agency *NNA* reported.

Israel said the strikes were a response to projectiles fired by the Iran-backed group at its troops in southern Lebanon, prompting attacks on what an Israeli official described as "Hezbollah targets".

A Hezbollah official told Reuters the group was committed to the ceasefire as long as Israel adheres to it, adding that Israeli forces do not have the right to freedom of movement in south Lebanon.

The official said Hezbollah forces have not fired on Israeli troops since 6:30 pm (1530 GMT) Saturday.

Security sources told Reuters Israel had also launched no major strikes on Lebanon since the same time.

That represents the longest spell without major strikes since the war in Lebanon between Hezbollah and Israel began on March 2.

When asked by *Reuters* about the last time Israel fired on Lebanon or engaged with Hezbollah, the Israeli military declined to provide a “specific time”. The Israeli military did not immediately respond to a request to comment on the claim they have ceased firing after 6:30 pm on Saturday.

Iranian officials have said Lebanon is the focal point for Sunday's peace talks with the United States in Switzerland after Washington and Tehran signed a framework to halt the war between them that began at the end of February, escalating tension across the region.

The Israeli military invaded parts of southern Lebanon. Hezbollah maintains it has the right to fight Israeli forces in Lebanon, but would halt attacks on northern Israel.

Katz said Israeli troops would remain in all positions in what it calls a security zone, which extends about 10 km (6 miles) into southern Lebanon.

Israel says that is to protect northern Israeli communities.

"All of the IDF's achievements in the campaign in Le-

A Lebanese army soldier at the site of an Israeli strike in Nabatieh, Lebanon, on Sunday. REUTERS

banon are being preserved," Katz said in Sunday's statement.

Israeli armed forces chief of staff Eyal Zamir said that the ceasefire in Lebanon was fragile and that forces must maintain a high level of readiness for a possible resumption of combat operations. Zamir said forces must be prepared to eliminate threats and transition rapidly to renewed operations if required.

Meanwhile Hezbollah chief Naim Qassem said Israel will not stay in Lebanon, adding that the group would respond to any violation from the Israeli side.

TIE-UP WITH HOLTEC INT’L ON THE CARDS, FOLLOWS PACTS WITH EDF, CLEAN CORE AND ROSATOM

Global nuclear players turn to NTPC for N-tech partnerships

It is the only state-owned company involved in construction of a nuclear power project, apart from NPCIL

Pratyush Deep & Anil Sasi
New Delhi, June 21

NTPC LTD, the country’s largest integrated power generator, is emerging as a key pivot in India’s nuclear landscape after the sector was opened up through an overarching legislation late last year.

For decades, the Nuclear Power Corporation of India Ltd (NPCIL) has been the vanguard of India’s traditional nuclear establishment. With the passage of the SHANTI Act, 2005, which eased the liability framework and paved the way for greater private and foreign participation, NTPC is emerging as a second pillar of sorts and a partner of choice for global nuclear technology providers seeking a foothold in the Indian market.

Notably, an NTPC functionary is also seen as a possible contender for a top energy advisory role in the government.

The latest collaboration in the works is with New Jersey-based nuclear technology firm

Holtec International known for its flagship small modular reactor, the SMR-300. The company is understood to be in advanced discussions with NTPC and could formalise a partnership as early as next month, *The Indian Express* has learned.

This comes after NTPC signed a non-binding Memorandum of Understanding (MoU) in April with Électricité de France (EDF) to explore co-operation in developing new nuclear power projects in India. The company has also inked an agreement with Russia’s state-owned nuclear company Rosatom for its VVER series of reactors — a type of reactors called light water reactors, or pressurised water reactor (PWR).

NTPC is also in talks to buy a minority stake in US-based nuclear fuel technology company Clean Core Thorium Energy (CCTE), the company had informed the stock exchange in January. Both the companies are exploring the “development and deployment of a thorium-based fuel” for

• DEEPENING NUCLEAR PUSH

NTPC’S LATEST collaboration is with New Jersey-based nuclear tech firm Holtec International known for its flagship small modular reactor, the SMR-300

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THE THERMAL major is also in talks to buy a minority stake in US-based nuclear fuel tech firm Clean Core Thorium Energy (CCTE)

CCTE AND NTPC are exploring the ‘development and deployment of a thorium-based fuel’ for existing nuclear stations in India

existing nuclear stations in India, subject to regulatory clearances, Department of Atomic Energy told the parliament earlier this year.

Both Holtec and Clean Core are among the handful of American nuclear companies which have begged ‘specific authorisations’ from the US government with respect to a restrictive American regulation referred to as “10CFR810” — (Part 810 of Title 10, Code of Federal Regulations of the US Atomic Energy Act of 1954).

The authorisation permits these firms, subject to certain

conditions, to transfer technology to Indian entities, which is otherwise barred under 810 regulations.

NTPC’s unique role, NPCIL’s hold on PHWRs

NTPC’s growing appeal among foreign nuclear technology providers stems from its unique position within India’s nuclear ecosystem. Apart from NPCIL, it is the only state-owned power company currently involved in the construction of a nuclear power project. NTPC is developing the Mahi Banswara Rajasthan Atomic

Power Project through ASHVINI, its joint venture with NPCIL.

In its latest earnings call, NTPC said it has secured capacity consent for 2,770 megawatt-electric (MWe) of the proposed 2,800 MWe Mahi Banswara Rajasthan Atomic Power Project from the customers in Rajasthan, Gujarat, Chhattisgarh, and Andhra Pradesh.

The first concrete pour for the project is expected by August 2027, while synchronisation of the first unit with the grid is targeted for November 2032, the company added.

Another factor working in favour of the state-owned thermal power major, as many in the policy circle point out, is NPCIL’s monopoly over the pressurised heavy water reactors (PHWR) technology which has long formed the backbone of India’s nuclear power programme.

While PHWRs are a technology India’s nuclear establishment has mastered, it is increasingly out of sync with the light water reactors (LWRs), or pressurised water reactors, that are now the most dominant type worldwide. The Americans, Russians and French are among the leaders in LWR technology.

FULL REPORT ON
WWW.INDIANEXPRESS.COM

USTR Greer to arrive today for trade talks, tariff edge vital: Goyal

ENS Economic Bureau
New Delhi, June 21

THE UNITED States Trade Representative (USTR) Jamie-son Greer will arrive here today to give final touches to the India-US interim trade agreement, Commerce and Industry Minister Piyush Goyal said on Sunday.

The ministerial-level discussion on the agreement between Goyal and Greer will be held on Tuesday and Wednesday. The talks follow two rounds of negotiations between Indian and US teams in the past two months.

Indian officials visited the US in April and in early June the negotiating team led by Assistant USTR Brendan Lynch was in India for four-day talks.

Last week, Prime Minister Narendra Modi met with US President Donald Trump for their first in-person talks in more than a year on the sidelines of the G7 summit in Evian, France. Both leaders struck an upbeat tone as they sought to turn the page on recent strains stemming from trade and other disputes, even as Trump said the deal has been “very close for a while.”

New Delhi is keen on a competitive advantage over countries competing with it in the US market, before rolling out the much-delayed trade agreement. “The issue currently pending is that our duties need to be lower compared to those of competing nations,” Goyal said. “Once this is settled, the trade agreement will be implemented.”

“The issue currently pending is that our duties need to be lower compared to those of competing nations. Once this is settled, the trade agreement will be implemented.”

PIYUSH GOYAL
COMMERCE & INDUSTRY MINISTER

He added that the framework for the pact was finalized before the US Supreme Court ruled in February that Trump’s previous tariff policy was unlawful.

The interim deal is based on the framework agreement reached between India and the US in February this year. As per the agreement, the US had agreed to bring down additional duties on Indian products to 18% from 50% in return for duty-free access for its manufactured products in Indian markets.

The agreement also mentions India buying \$500 billion worth of goods from the US over five years.

While the reciprocal tariffs, which sparked the trade negotiations, are no longer in force, the US is building another framework of tariffs using Section 301 of the Trade Act to enter into trade deals with partners. India’s contention is that it must get lower tariffs among its competitors in the US market. FE

NEWSMAKERS IN THE NEWSROOM

WHY DENIS ALIPOV

Denis Alipov has been the Russian ambassador to India for more than four years, coming here in February 2022, just before the war in Ukraine began. He has lived in India for a few years as a child, went to Jawaharlal Nehru University to study for a year, and has done three diplomatic postings here. This is his fourth posting. Alipov is considered one of the top India experts within the Russian government. He had also worked with Russian Foreign minister Sergei Lavrov in his office. He is best placed to make sense of Russia-India ties at a time of political turbulence and the recent developments in geopolitics from that vantage point



ON INDIA'S RUSSIAN OIL IMPORTS

'Russian oil has been advantageous for India over the past several years. We continue to offer competitive prices. If India chooses to reduce imports from Russia, we will simply redirect those supplies to other markets'



ON INDIANS IN RUSSIAN MILITARY

'At present, there are no Indian citizens serving in the Russian armed forces. We stopped accepting Indian volunteers approximately two years ago following a request from the Indian government'

'We will never allow China to affect our ties with India, which remain a high priority in our foreign policy'

Denis Alipov, the Russian Ambassador to India, on the solid foundation of the India-Russia relationship, the impact of Western sanctions on defence exports and its continued conflict with Ukraine. He was in conversation with Shubhajit Roy, Diplomatic Editor, *The Indian Express*



Russian Ambassador Denis Alipov (right) with Shubhajit Roy, Diplomatic Editor, *The Indian Express* RENUKA PURI

manitarian exchanges and labour mobility.

Ritu Sarin: Could you provide an update on two aspects of India–Russia trade relations? First, the contract for 120 Vande Bharat trains that Russia is expected to supply, and second, the delay in Gazprombank commencing operations in India.

We have been awaiting the issuance of a licence for Gazprombank by the Reserve Bank of India. The delay is not on the Russian side. More broadly, we are keen to expand our financial relationship, including by increasing the presence of Russian banks in India and encouraging more Indian banks to operate in Russia. As for the Vande Bharat project, we won the contract and the work is proceeding according to schedule. The first prototype is expected by the end of this year.

Anil Sasi: Has the conflict in West Asia exposed the limitations of American security guarantees? And does that, in any way, overlap with Russia's concerns about NATO expansion?

NATO is, in our view, an outdated organisation that should be dismantled. Our assessment is that the United States has failed as a security provider for the Gulf countries and made a serious mistake by attacking Iran. We also observe a broader shift in global attitudes towards the United States. Its position as the sole dominant global power is weakening. President Trump's current emphasis on domestic priorities suggests a reduced interest in maintaining claims to global dominance which, in our view, would be beneficial for the international system.

Shubhajit Roy: Many would argue that Iran has been one of Russia's key partners, particularly given its supply of Shahed drones. Russia is, therefore, also a significant actor in the region. Is the current situation a test case of

whether Russia can push back against American influence as a security guarantor in the Middle East?

We have no intention of imposing ourselves on the Middle East or replacing any other external power in the region. However, we do have legitimate interests there and we are determined to defend them.

Russia has long been a member of the Quartet on the Middle East peace process, which has regrettably been sidelined in recent years, particularly following the emphasis placed on the Abraham Accords. We continue to believe that the Palestinian issue can only be resolved on the basis of a two-state solution. The region to Russia's south is of considerable strategic importance to us. I would not say that we have been as actively involved in efforts to end the conflict involving Iran as countries such as Pakistan, Qatar and several other Gulf states. Nevertheless,

we have made proposals to both Washington and Tehran and offered assistance, including with regard to the storage of enriched uranium. Those proposals were not accepted. President Putin has discussed the possibility of Russian involvement in resolving the crisis with both President Trump and the Iranian leadership. Russia remains ready to assist whenever our help is required.

Anil Sasi: How do you explain Ukraine's ability to reclaim territory and conduct strikes deep inside Russia?

Ukraine has intensified its drone attacks, and we have responded in kind. On the battlefield, Russia is advancing gradually but steadily across all sectors of the front line. The basic reality is that the defeat of Russia, on which Ukraine and some European governments continue to place their hopes, is

ON UKRAINE'S STRIKES IN RUSSIA

'We have responded in kind. On the battlefield, Russia is advancing gradually but steadily across all sectors of the front line. The defeat of Russia, on which Ukraine and Europe continue to place their hopes, is objectively impossible'

objectively impossible.

Europe's attempt to wear Russia down economically has also failed. We possess a significantly higher threshold of endurance. Europe is not prepared to put everything at stake, whereas Russia is. It should also not be forgotten that Russia is a nuclear power. We retain the right to use nuclear weapons if the existence or territorial integrity of the Russian state is threatened.

P Vaidyanathan Iyer: European economic pressure has pushed Russia closer to China. How do you view this growing reliance?

Our trade with China was expanding well before 2022. At the same time, we remain determined to deepen our economic relationship with India, although the response from the Indian side has been more cautious. The assumption that Russia's relationship with China should automatically be viewed as a sensitive issue for India is fundamentally mistaken. A broader and more diversified economic partnership between Russia and India would help alleviate such concerns.

P Vaidyanathan Iyer: Do you see significant reluctance on India's part when it comes to expanding trade with Russia?

Not on the part of the government. The business and financial communities have naturally been more sensitive to the impact of sanctions, and we understand that. India's economy is deeply integrated into global trade and financial networks. India would not wish to sacrifice its relations with the United States and Europe for the sake of its relationship with Russia. My point is that Western sanctions interfere in what should be a bilateral relationship between Russia and India. These are matters that do not concern third parties. We see this as a manifestation of a neo-colonial mindset — the notion that others know better than India what

India's interests should be. Such pressure tactics are illegitimate and should be rejected.

Flora Swain: There have been reports of Indian nationals and North Korean soldiers participating in the war in Ukraine on Russia's side. What safeguards are in place to prevent foreign nationals from being recruited into the Russian military?

Russia has faced demographic challenges for more than three decades. That is a long-term issue and is unrelated to the conflict in Ukraine. At present, there are no Indian citizens serving in the Russian armed forces. We stopped accepting Indian volunteers approximately two years ago following a request from the Indian government.

In fact, Russia has never operated a programme specifically designed to recruit foreign nationals. Those foreign citizens who serve in the Russian military do so on a voluntary basis.

Sukalp Sharma: Given the uncertainty surrounding sanctions and India's purchases of Russian oil, do you expect current import levels to be maintained once the latest US sanctions waiver expires? And how do you assess India's continued purchases of Russian crude following sanctions imposed on companies such as Rosneft and Lukoil?

I cannot predict the precise volumes of Russian oil that India will purchase in the future, either with sanctions in place or without them. What is clear is that Russian oil has been economically advantageous for India over the past several years, particularly after Europe increased its purchases of Middle Eastern crude, which had previously been one of India's principal sources of supply. We continue to offer oil at competitive prices, and India will always be welcome to purchase it. If India chooses to reduce imports from Russia, we will simply redirect those supplies to other markets. We are comfortable with either outcome. Nevertheless, Russian oil remains among the most competitively priced options available. As of May this year, Russia accounted for approximately 38 per cent of India's total crude oil imports.

Shubhajit Roy: Before that, Russia accounted for less than one per cent of India's imports.

Sanctions should be understood for what they are: a political instrument of pressure. They should be rejected.

There is another, less obvious but equally important point. Even if a political settlement in Ukraine is eventually reached, sanctions may remain in place for a considerable period. Waiting for them to disappear would only damage Russia–India cooperation over the long term. Neither Russia nor India wants that outcome. Russia remains, and will continue to remain, one of India's principal partners. Nuclear energy, defence and space cooperation are among the key pillars of our relationship.

Rosatom is the world's leading nuclear energy company. We have worked successfully with India for many years, even without legislation such as the SHANTI Act, while fully accommodating India's positions on safety, liability and regulatory conditions.

We are also engaged in extensive discussions on artificial intelligence and large language models. Russia and India cannot rely exclusively on technologies developed in either the United States or China, and both countries are investing heavily in developing their own capabilities.

Sukalp Sharma: G7 sanctions and the oil price cap have led to what the West calls Russia's 'shadow fleet'. Critics have concerns over their age, safety standards, the disabling of AIS transponders and ship-to-ship transfers. How do you respond?

These are artificial pretexts. The term 'shadow fleet' is used because many of these vessels are not insured through Western providers such as Lloyd's. They are insured by Russian, Indian and other non-Western insurers and should, therefore, be regarded as part of a normal commercial fleet. This is yet another example of double standards being applied in order to single out political opponents. The actions recently taken by Britain and France are unacceptable. Russia has begun escorting certain vessels and tankers with naval ships in order to prevent such incidents. The vessel detained recently in the English Channel was not sailing under the Russian flag. It may have been carrying Russian oil, although I cannot say whether the cargo was destined for India. In our view, such actions amount to a form of piracy that Britain appears to have revived.

Ravi Dutta Mishra: What is the current status of rupee–rouble trade and Sberbank's operations in India?

We already have a functioning mechanism for conducting transactions in national currencies — roubles and rupees — and Sberbank is making full use of it. Indian banks should also take greater advantage of these opportunities and of the bilateral mechanisms that have been established to expand trade in national currencies.

Sberbank has maintained an active presence in India and continues to expand its activities. It is also involved in projects related to artificial intelligence, the digital economy, digital solutions and education.



CONSULTATION ROOM

DR AMBUJ ROY

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Can yoga complement medicine? The evidence is growing

Studies are showing results, from heart attack recovery to prevention

EVERY YEAR on International Yoga Day, millions roll out their mats, celebrating a practice that has been part of India's civilisational heritage for millennia. Yet the most important development of yoga for us in academia is happening beyond parks or stadiums. It is happening in research laboratories, clinical trials and medical journals.

Yoga is steadily moving from a belief system to an evidence-based health intervention. The numbers tell a compelling story. Nearly 90 per cent of all scientific publications on yoga in medical literature have appeared in the last two decades. In fact, this has accelerated further in the last decade, with the number of yoga-related clinical trials reported in PubMed (the leading biomedical search engine) more than doubling in the most recent decade compared with the preceding 10 years.

This transition is particularly important in my field of cardiovascular medicine, where claims are scrutinised rigorously and only interventions supported by evidence find a place in clinical practice. Today, that evidence is beginning to emerge and become mainstream. One of the landmark studies in this field was the Indian Council of Medical Research-funded trial by our Yoga-CaRe group of authors, published in the *Journal of the American College of Cardiology* in 2020. Conducted across 24 centres in India and among nearly 4,000 patients recovering from heart attacks, it remains the largest randomised clinical trial ever undertaken on yoga-based cardiac rehabilitation. It found that patients who participated in a structured yoga-based rehabilitation programme reported significantly better health status and earlier return to normal daily activities, following a heart attack. In fact, among practitioners who attended 75% or more of yoga sessions, there were 40% fewer adverse cardiovascular outcomes. The Yoga-CaRe trial was important not merely for its findings but also for demonstrating that yoga can be evaluated using the same rigorous standards applied to medicines and medical devices.

More recently, CaReMATCH, an analysis of several cardiac rehabilitation trials, showed that yoga was as beneficial as conventional cardiac rehabilitation in improving quality of life and preventing recurrent hospitalisations by 30%. This has important implications for India, since the availability and acceptability of standard cardiac rehabilitation with multiple instruments and gadgets is limited and expensive, while yoga is culturally acceptable and readily available and affordable.

A yoga-based rehabilitation model can extend the reach of cardiovascular care far beyond tertiary hospitals into smaller cities and rural communities. Research suggests that yoga works by improving autonomic balance (equilibrium between the sympathetic fight-or-flight and the parasympathetic rest-and-digest systems) enhancing heart-rate variability (the natural fluctuation in the exact time intervals between consecutive heartbeats and an indicator of how well your body adapts to stress and recovers from exertion) reducing stress-related hormonal activation, and improving psychological well-being. These mechanisms are key to cardiovascular health.

Evidence for yoga is emerging not only in cardiac rehabilitation but also in preventive cardiovascular health. A growing body of research suggests that regular yoga practice can improve blood pressure, body weight, lipid levels and, importantly, quality of life. A pooled analysis of 49 trials found that yoga produced average reductions of 5 mmHg in systolic and 3 mmHg in diastolic blood pressure. Such reductions can lead to 25% and 20% lowering in the risk of brain and heart attack, respectively. The public health implications of this can be profound. Even modest improvements in these risk factors can translate into large reductions in cardiovascular disease and other non-communicable conditions, including disorders affecting brain health, kidney disease and bone health, when adopted across large populations.

Beyond disease prevention, yoga promotes positive health and has been shown to improve physical fitness, mental well-being, emotional resilience, sleep quality and healthy ageing — outcomes that are increasingly recognised as important components of overall health.

The next step is to expand access to yoga while systematically measuring its impact on health outcomes. Doing so will help generate the evidence needed to guide policy and maximise yoga's contribution to public health. Positive results should lead to its systematic implementation through the extensive network of Ayushman Aarogya Mandirs as a proactive health policy. The real story of yoga in the 21st century is, therefore, not one of ancient wisdom competing with modern science. It is one of ancient wisdom being examined, tested and, where justified, validated by modern science. India has long exported yoga as culture. The moment has come to export it as evidence-based therapy.

Over 2,000 cough syrup brands: Can regulators ensure they're safe?

Cough syrups are among the most consumed medicines but recurring contamination incidents have exposed flaws in quality checks

Anonna Dutt

COUGH SYRUPS are among the most widely manufactured and consumed medicines in India, making robust regulation of the market essential. In May alone, sales touched at least 8.47 lakh units worth Rs 6,670 crore, according to pharma retail tracker Pharmarack. Yet the sector has repeatedly come under scrutiny after contaminated cough syrups were linked to the deaths of children in India and abroad. The Health Ministry has since tightened rules, restricting sales to licenced chemists and making doctors' prescription mandatory.

With more than 2,000 manufacturers producing cough syrups across the country, experts say sustained regulatory vigil, rigorous quality testing and stricter enforcement are critical to safeguarding public health. "Cough syrup manufacturers are like *halwais* — they are everywhere. This is the reason it is very difficult to maintain the quality of the products. While bigger companies may implement good manufacturing practices, ensure sterile water and environment, it is not true for most manufacturers," says an industry expert, who did not wish to be named.

This is a cause for concern, considering the apex drug regulator did not have on its record the Tamil Nadu-based firm whose cough syrup was implicated in the deaths of at least 22 children in Madhya Pradesh (MP) last year. After the incident, the drug regulator admitted that the company was not on its records and that the state had never shared information about them with the Central Drugs Standard Control Organisation (CDSCO).

Massive inspection drive but does it yield results?

Following the incident, the apex regulator reached out to all states to gather data on all cough syrup manufacturers across the country and then carry out a risk-based inspection — vetting of the premises of a manufacturing unit that has been repeatedly flagged for substandard drugs.

The CDSCO, along with state regulators, has inspected the premises of 960 companies since last year, based on which 860



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actions were taken, including issuance of showcase notices, stop production order, suspension, cancellation of licences / product licences and warning letters. Around 1,100 manufacturers have been subjected to intense audit since.

"While the states were asked to share details of all cough syrup manufacturers after the incident in MP, the challenge is that some may still not do so," says an official in the know of the matter. Another expert on drug regulation adds: "If the company is not on record itself, how can risk-based inspection be carried out? For risk-based inspection, data of previous regulatory action and instances of samples failing are used to determine which companies or manufacturing units need to be inspected."

Where are manufacturing clusters located?

Manufacturing units of the companies making the most common cough products — by sale and by volume — are largely clustered in states such as Maharashtra, Himachal Pradesh, Sikkim, Gujarat, Madhya Pradesh and Goa.

Some of the topmost common brands of cough syrups sold in India are Cipla's Corex, Glenmark's Ascorig, Centaur's Sinarest, Dr Reddy's Zedex and Johnson and Johnson's Benadryl among others.

There's need for a consolidated database across states and the Centre for regulatory information



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What are the major regulatory challenges?

While the current move by the Centre is likely to have limited impact, experts say that ensuring availability of quality raw materials — even to the smaller companies — is likely to reduce instances of contamination. "The 2,000 manufacturers of cough syrup are not the problem, the fragmented regulatory ecosystem in India is. India today has effectively 37 regulators, one per state/UT and the national regulator — the CDSCO. There is no consolidated database containing accurate information from all these 37 regulators, despite India being a single market for all practical purposes with drugs being able to cross state borders as easily as any other commodity," says public health activist Dinesh Thakur.

So, a drug manufactured in West Bengal can be sold in Karnataka but the Karnataka drug controller cannot inspect the plant in West Bengal or cancel its manufacturing licence if it sells substandard drugs in Karnataka. "At most the Karnataka drug controller can launch a criminal prosecution and that can take a lot of time in Indian courts. Even if there is enough evidence, the courts tend to be very lenient. It is this fragmented regulatory framework that is the problem. Despite petitions, the core issue is still unaddressed," adds Thakur.

Why do contaminants enter the pharma supply chain?

Pharmaceutical companies have not been testing their ingredients, especially propylene glycol, the solvent they use to manufacture cough syrups. Besides, these drugs are difficult to manufacture because of sterility requirements. "This is a brazen violation of the GMP (good manufacturing practices) code. Given the lax nature of regulation in India, most of these companies know they can escape punitive consequences," says Thakur.

An industry expert concurs: "Smaller companies have to buy propylene glycol from the open market. Unlike bigger companies, they cannot afford to buy full, sealed barrels from suppliers. This also creates an opportunity for the unscrupulous people to mix cheap industrial grade propylene glycol as pharmaceutical grade."

Thakur argues for a consolidated database of licensing, inspection, marketing information across states and the Centre to ensure a single source of truth of all regulatory information. "This is not just meant for regulatory purposes but also procurement purposes by both the public and private sector," he adds.

He believes the industry's argument that propylene glycol has to be regulated doesn't quite hold. "No other country regulates raw materials. It is the responsibility of the pharma company, as it is written in the law, to test their raw materials before using them. At the end of the day, it all boils down to compliance with the GMP code — if followed properly, all these issues will be caught before the product is shipped to the market. The Indian pharmaceutical industry does not care because manufacturers know they will escape any strict action from the state as evidenced by the outcome of so many tragedies in the recent past. Can you point one instance where a particular company or its management has been punitively held to account?," asks Thakur.

The cough syrup tragedies are only the tip of the iceberg; they are evidence of how a regulatory oversight can jeopardise the manufacturer's credibility. "Cancer medication manufactured by Indian companies has been found to be contaminated in Colombia and Yemen. There are a huge number of complaints coming in from Nepal, Sri Lanka and many African countries. Poorly manufactured eye drops have made their way to the US. Then there are doctors who have discovered that their patients often do not react as expected to Indian manufactured drugs. The Indian industry's reputation is taking a beating across the globe — at some point when more trustworthy competitors emerge, it will be the end of the Indian pharmaceutical dream," says Thakur.

(With Rinku Ghosh)

● DECODING A STUDY

IIT Bombay finds a new way to lower LDL cholesterol

Researchers stop bad fat from reaching the site inside liver cells that releases it into the blood

Purnima Sah

FOR DECADES, scientists have tried to lower harmful fats in the blood by targeting enzymes, genes or receptors involved in fat metabolism. Researchers at IIT Bombay have now taken a strikingly different approach: preventing bad fat from reaching the site inside liver cells that releases it into the bloodstream.

Working with scientists at IISER (Indian Institute of Science Education and Research) Pune and IISER Kolkata, the team has developed a tiny peptide (a short chain of amino acids or the building blocks of life) called KTDP that interrupts the movement of fat inside liver cells. In laboratory-grown liver cells and zebrafish, this reduced the release of cholesterol and triglycerides into the bloodstream by nearly 50 per cent.

The findings, published recently in the *Proceedings of the National Academy of Sciences (PNAS)*, are still at an early stage. The peptide has not been tested in humans. But researchers say the work opens an en-

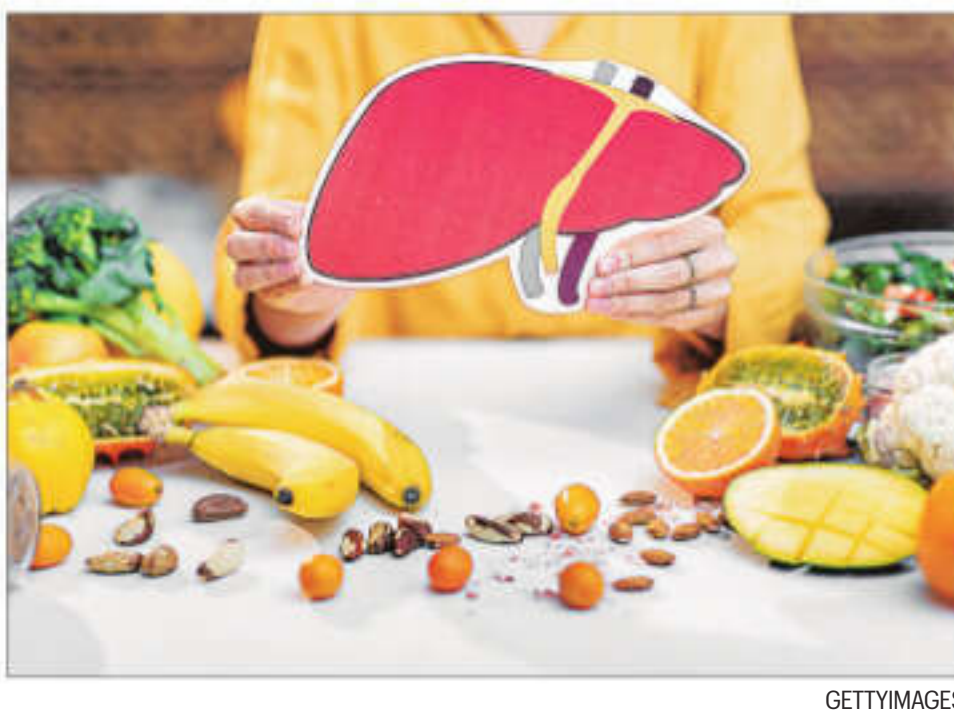
tirely new strategy for reducing the biggest risk factor of heart attacks, namely bad fats — LDL (low-density lipoprotein) and triglycerides — in the blood.

Why does this finding matter?

While medicines such as statins have transformed the treatment of high cholesterol, options for lowering triglycerides remain relatively limited. The IIT Bombay study proposes a fresh way of reducing the liver's export of fats into the bloodstream rather than interfering with how fats are produced or broken down.

The liver is the body's central hub for fat metabolism. Inside liver cells are tiny storage compartments called lipid droplets that temporarily hold fats such as cholesterol and triglycerides. When required, these droplets are transported to specific locations within the cell, where they are repackaged into very low-density lipoproteins (VLDL) and released into the bloodstream.

Professor Roop Mallik of IIT Bombay explains it simply. "There are little balls of fat



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Blocking fat export did not lead to dangerous fat build-up inside the liver. Fatty acids were instead redirected to the mitochondria

inside our liver cells. These need to be carried to specific locations inside the liver cell. From there, the fat is repackaged into lipoprotein particles and sent out into your blood," he says. "If we can tone down the export of fat from the liver into blood using a small peptide, we could have a new way of treating high serum lipids." The researchers' solution was KTDP, a tiny peptide derived from a naturally occurring protein.

What is KTDP?

Researchers identified a motor protein

called kinesin-1, which is a delivery truck operating inside a cell. "These motor proteins catch hold of these fat particles and take them to a certain location. We found a small piece of this protein called a peptide. When we delivered this to cells or zebrafish, these didn't get carried to the right location," Prof Mallik says. Without reaching that destination, the fat cannot be packaged into VLDL particles and exported into the bloodstream.

How is this different from statins?

Current cholesterol-lowering medicines, including statins, work by reducing cholesterol synthesis or increasing its clearance from the blood but have nothing to halt it in its tracks. Because the peptide selectively blocks the transport of lipid droplets without shutting down protein functions overall, researchers believe it could avoid some of the side-effects associated with disrupting a motor protein.

Excess fat circulating in the blood can accumulate in several organs. "That is a disaster because that fat is going to accumulate in your heart and other tissues," says Prof Mallik. Interestingly, blocking fat export did not lead to dangerous fat build-up inside the liver. Fatty acids were instead redirected to the mitochondria — the cell's energy-producing structures.

